Social cue means and distribution shape in prediction of pain and visual ratings

Exploratory secondary analysis of public behavioural data

DATA AND VALIDATION

45 participants

6,035 retained trials

Thermal pain + visual contrast

Train on 44 people; test on 1

45 leave-one-participant-out folds per modality

PARTICIPANT-WEIGHTED RMSE

Lower is better; rating-point units

	M0 Stimulus + time	M1 + Cue mean	M2 + SD + skewness
Pain	22.467	21.660	21.669
Vision	16.352	15.623	15.614

Adding SD and skewness changed RMSE by less than 0.01 points in either modality.

The model including cue mean recovered nearly all the modest gain

Additional distribution features offered little gain in these fixed linear models.

One experiment | Exploratory comparison | No external validation or mechanistic inference

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Data: Botvinik-Nezer et al. (2025) | PPRI-paper v2.0.0